

Technical Debt Register & Phased Repayment Plan

Project Title: SoluoPrint - Premium All-in-One Print Shop Management (Print/Job management, CRM, Financial Management & Analytics Platform, Automated SMS Notifications)

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NOTE: Technical Debt Philosophy

In a 48-hour development timeframe, strategic technical debt is deliberately incurred to prioritize core commercial requirements. All debt items are cataloged, prioritized, and assigned a remediation roadmap.

1. Technical Debt Register

Debt ID	Technical Debt Item	Root Cause Analysis	Architectural Impact	Priority Rating	Resolution Status / Classification
TD-01	Manual Password Hashing in Prototype Auth	Rapid prototyping shortcut	High Security Risk	CRITICAL	RESOLVED (Migrated to Supabase Auth RLS) - Immediate Attention
TD-02	Lack of Automated Playwright/Cypress E2E Tests	48-Hour Exam Time Constraint	Regression risk on future updates	HIGH	Scheduled for Release v1.1 - Scheduled for Future
TD-03	Monolithic Dashboard Component	Speed of initial UI assembly	Decreased code maintainability	MEDIUM	Scheduled for Release v1.1 - Acceptable Temporarily
TD-04	Hardcoded API Gateway Secrets	Quick API verification during exam	Security risk in repository	HIGH	RESOLVED (Extracted to cPanel & .env) - Immediate Attention
TD-05	Client-Side Financial	Reduced backend RPC	Increased memory	MEDIUM	Scheduled for Release v2.0 -

	Aggregations	function overhead	footprint for large datasets		Acceptable Temporarily
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2. Phased Technical Debt Repayment Roadmap

- **Phase 1 (Immediate - Resolved):** Fixed TD-01 and TD-04 by enforcing Supabase Auth Row Level Security (RLS) policies and extracting API credentials into environment variables on Namecheap cPanel. These items required immediate attention due to security implications.
- **Phase 2 (Short-Term - Release v1.1):** Implement Playwright E2E automated test suite (TD-02) and refactor DashboardPage.jsx into modular widgets (TD-03) to ensure maintainability as new features are introduced.
- **Phase 3 (Medium-Term - Release v2.0):** Migrate heavy financial report math from client JavaScript to Supabase PostgreSQL RPC stored procedures (TD-05) to optimize performance for large datasets.